What does this checklist tell you about your mastery of this section? What steps will you take to improve?

5.4 Solve Equations with Decimals

Learning Objectives

By the end of this section, you will be able to:

- Determine whether a decimal is a solution of an equation
- Solve equations with decimals
- Translate to an equation and solve

BE PREPARED 5.10 Before you get started, take this readiness quiz.

Evaluate $x + \frac{2}{3}$ when $x = -\frac{1}{4}$.

If you missed this problem, review Example 4.77.

BE PREPARED 5.11 Evaluate $15 - y$ when $y = -5$.

If you missed this problem, review Example 3.41.

BE PREPARED 5.12 Solve $\frac{n}{-7} = 42$.

If you missed this problem, review Example 4.99.

Determine Whether a Decimal is a Solution of an Equation

Solving equations with decimals is important in our everyday lives because money is usually written with decimals. When applications involve money, such as shopping for yourself, making your family's budget, or planning for the future of your business, you'll be solving equations with decimals.

Now that we've worked with decimals, we are ready to find solutions to equations involving decimals. The steps we take to determine whether a number is a solution to an equation are the same whether the solution is a whole number, an integer, a fraction, or a decimal. We'll list these steps here again for easy reference.

HOW TO

Determine whether a number is a solution to an equation.

Step 1. Substitute the number for the variable in the equation.
Step 2. Simplify the expressions on both sides of the equation.
Step 3. Determine whether the resulting equation is true.
- If so, the number is a solution.
- If not, the number is not a solution.

EXAMPLE 5.40

Determine whether each of the following is a solution of $x - 0.7 = 1.5$:

(a) $x = 1$ (b) $x = -0.8$ (c) $x = 2.2$

Solution

(a)

	$x - 0.7 = 1.5$

Substitute 1 for x.	$1 - 0.7 \stackrel{?}{=} 1.5$
Subtract.	$0.3 \neq 1.5$

Since $x = 1$ does not result in a true equation, 1 is not a solution to the equation.

ⓑ

	$x - 0.7 = 1.5$
Substitute -0.8 for x.	$-0.8 - 0.7 \stackrel{?}{=} 1.5$
Subtract.	$-1.5 \neq 1.5$

Since $x = -0.8$ does not result in a true equation, -0.8 is not a solution to the equation.

ⓒ

	$x - 0.7 = 1.5$
Substitute 2.2 for x.	$2.2 - 0.7 \stackrel{?}{=} 1.5$
Subtract.	$1.5 = 1.5$ ✓

Since $x = 2.2$ results in a true equation, 2.2 is a solution to the equation.

TRY IT 5.79 Determine whether each value is a solution of the given equation.

$x - 0.6 = 1.3$: ⓐ $x = 0.7$ ⓑ $x = 1.9$ ⓒ $x = -0.7$

TRY IT 5.80 Determine whether each value is a solution of the given equation.

$y - 0.4 = 1.7$: ⓐ $y = 2.1$ ⓑ $y = 1.3$ ⓒ -1.3

Solve Equations with Decimals

In previous chapters, we solved equations using the Properties of Equality. We will use these same properties to solve equations with decimals.

Properties of Equality

Subtraction Property of Equality For any numbers $a, b,$ and c, If $a = b$, then $a - c = b - c$.	**Addition Property of Equality** For any numbers $a, b,$ and c, If $a = b$, then $a + c = b + c$.
The Division Property of Equality For any numbers $a, b,$ and $c,$ and $c \neq 0$ If $a = b$, then $\frac{a}{c} = \frac{b}{c}$	**The Multiplication Property of Equality** For any numbers $a, b,$ and c, If $a = b$, then $ac = bc$

When you add, subtract, multiply or divide the same quantity from both sides of an equation, you still have equality.

EXAMPLE 5.41

Solve: $y + 2.3 = -4.7$.

Solution

We will use the Subtraction Property of Equality to isolate the variable.

	$y + 2.3 = -4.7$
Subtract 2.3 from each side, to undo the addition.	$y + 2.3 - 2.3 = -4.7 - 2.3$
Simplify.	$y = -7$

Check:	$y + 2.3 = -4.7$
Substitute $y = -7$.	$-7 + 2.3 \stackrel{?}{=} -4.7$
Simplify.	$-4.7 = -4.7$ ✓

Since $y = -7$ makes $y + 2.3 = -4.7$ a true statement, we know we have found a solution to this equation.

TRY IT 5.81 Solve: $y + 2.7 = -5.3$.

TRY IT 5.82 Solve: $y + 3.6 = -4.8$.

EXAMPLE 5.42

Solve: $a - 4.75 = -1.39$.

Solution

We will use the Addition Property of Equality.

	$a - 4.75 = -1.39$
Add 4.75 to each side, to undo the subtraction.	$a - 4.75 + 4.75 = -1.39 + 4.75$
Simplify.	$a = 3.36$

Check:	$a - 4.75 = -1.39$
Substitute $a = 3.36$.	$3.36 - 4.75 \stackrel{?}{=} -1.39$
	$-1.39 = -1.39$ ✓

Since the result is a true statement, $a = 3.36$ is a solution to the equation.

TRY IT 5.83 Solve: $a - 3.93 = -2.86$.

TRY IT 5.84 Solve: $n - 3.47 = -2.64$.

EXAMPLE 5.43

Solve: $-4.8 = 0.8n$.

Solution

We will use the Division Property of Equality.

Use the Properties of Equality to find a value for n.

	$-4.8 = 0.8n$
We must divide both sides by 0.8 to isolate n.	$\frac{-4.8}{0.8} = \frac{0.8n}{0.8}$
Simplify.	$-6 = n$

Check:	$-4.8 = 0.8n$
Substitute $n = -6$.	$-4.8 \stackrel{?}{=} 0.8(-6)$
	$-4.8 = -4.8$ ✓

Since $n = -6$ makes $-4.8 = 0.8n$ a true statement, we know we have a solution.

TRY IT 5.85 Solve: $-8.4 = 0.7b$.

TRY IT 5.86 Solve: $-5.6 = 0.7c$.

EXAMPLE 5.44

Solve: $\frac{p}{-1.8} = -6.5$.

Solution

We will use the Multiplication Property of Equality.

	$\frac{p}{-1.8} = -6.5$
Here, p is divided by −1.8. We must multiply by −1.8 to isolate p	$-1.8\left(\frac{p}{-1.8}\right) = -1.8(-6.5)$
Multiply.	$p = 11.7$

Check:	$\frac{p}{-1.8} = -6.5$
Substitute $p = 11.7$.	$\frac{11.7}{-1.8} \stackrel{?}{=} -6.5$
	$-6.5 = -6.5$ ✓

A solution to $\frac{p}{-1.8} = -6.5$ is $p = 11.7$.

TRY IT 5.87 Solve: $\frac{c}{-2.6} = -4.5$.

TRY IT 5.88 Solve: $\frac{b}{-1.2} = -5.4$.

Translate to an Equation and Solve

Now that we have solved equations with decimals, we are ready to translate word sentences to equations and solve. Remember to look for words and phrases that indicate the operations to use.

EXAMPLE 5.45

Translate and solve: The difference of n and 4.3 is 2.1.

Solution

Translate.	The difference of n and 4.3 is 2.1. $n - 4.3 = 2.1$
Add 4.3 to both sides of the equation.	$n - 4.3 + 4.3 = 2.1 + 4.3$
Simplify.	$n = 6.4$

Check:	Is the difference of n and 4.3 equal to 2.1?
Let $n = 6.4$:	Is the difference of 6.4 and 4.3 equal to 2.1?

Translate.	$6.4 - 4.3 \stackrel{?}{=} 2.1$	
Simplify.	$2.1 = 2.1$ ✓	

TRY IT 5.89 Translate and solve: The difference of y and 4.9 is 2.8.

TRY IT 5.90 Translate and solve: The difference of z and 5.7 is 3.4.

EXAMPLE 5.46

Translate and solve: The product of -3.1 and x is 5.27.

Solution

Translate.		$\underbrace{\text{The product of } 3.1 \text{ and } x}_{-3.1x} \quad \underbrace{\text{is}}_{=} \quad \underbrace{5.27.}_{5.27}$
Divide both sides by -3.1.		$\frac{-3.1x}{-3.1} = \frac{5.27}{-3.1}$
Simplify.		$x = -1.7$
Check:	Is the product of -3.1 and x equal to 5.27?	
Let $x = -1.7$:	Is the product of -3.1 and -1.7 equal to 5.27?	
Translate.	$-3.1(-1.7) \stackrel{?}{=} 5.27$	
Simplify.	$5.27 = 5.27$ ✓	

TRY IT 5.91 Translate and solve: The product of -4.3 and x is 12.04.

TRY IT 5.92 Translate and solve: The product of -3.1 and m is 26.66.

EXAMPLE 5.47

Translate and solve: The quotient of p and -2.4 is 6.5.

Solution

Translate.	The quotient of p and -2.4 is 6.5. $\frac{p}{-2.4} = 6.5$
Multiply both sides by -2.4.	$-2.4\left(\frac{p}{-2.4}\right) = -2.4(6.5)$
Simplify.	$p = -15.6$

Check: Is the quotient of p and -2.4 equal to 6.5?

Let $p = -15.6$: Is the quotient of -15.6 and -2.4 equal to 6.5?

Translate.	$\frac{-15.6}{-2.4} \stackrel{?}{=} 6.5$
Simplify.	$6.5 = 6.5$ ✓

TRY IT 5.93 Translate and solve: The quotient of q and -3.4 is 4.5.

TRY IT 5.94 Translate and solve: The quotient of r and -2.6 is 2.5.

EXAMPLE 5.48

Translate and solve: The sum of n and 2.9 is 1.7.

Solution

Translate.	The sum of n and 2.9 is 1.7. $n + 2.9 = 1.7$
Subtract 2.9 from each side.	$n + 2.9 - 2.9 = 1.7 - 2.9$
Simplify.	$n = -1.2$

Check: Is the sum n and 2.9 equal to 1.7?

Let $n = -1.2$: Is the sum -1.2 and 2.9 equal to 1.7?

Translate.	$-1.2 + 2.9 \stackrel{?}{=} 1.7$
Simplify.	$1.7 = 1.7$ ✓

TRY IT 5.95 Translate and solve: The sum of j and 3.8 is 2.6.

TRY IT 5.96 Translate and solve: The sum of k and 4.7 is 0.3.

MEDIA

ACCESS ADDITIONAL ONLINE RESOURCES

Solving One Step Equations Involving Decimals (https://openstaxcollege.org/l/24eqwithdec)
Solve a One Step Equation With Decimals by Adding and Subtracting (https://openstaxcollege.org/l/24eqnwdecplsmin)
Solve a One Step Equation With Decimals by Multiplying (https://openstaxcollege.org/l/24eqnwdecmult)
Solve a One Step Equation With Decimals by Dividing (https://openstaxcollege.org/l/24eqnwdecdiv)

SECTION 5.4 EXERCISES

Practice Makes Perfect

Determine Whether a Decimal is a Solution of an Equation

In the following exercises, determine whether each number is a solution of the given equation.

285. $x - 0.8 = 2.3$
ⓐ $x = 2$ ⓑ $x = -1.5$ ⓒ $x = 3.1$

286. $y + 0.6 = -3.4$
ⓐ $y = -4$ ⓑ $y = -2.8$ ⓒ $y = 2.6$

287. $\frac{h}{1.5} = -4.3$
ⓐ $h = 6.45$ ⓑ $h = -6.45$ ⓒ $h = -2.1$

288. $0.75k = -3.6$
ⓐ $k = -0.48$ ⓑ $k = -4.8$ ⓒ $k = -2.7$

Solve Equations with Decimals

In the following exercises, solve the equation.

289. $y + 2.9 = 5.7$

290. $m + 4.6 = 6.5$

291. $f + 3.45 = 2.6$

292. $h + 4.37 = 3.5$

293. $a + 6.2 = -1.7$

294. $b + 5.8 = -2.3$

295. $c + 1.15 = -3.5$

296. $d + 2.35 = -4.8$

297. $n - 2.6 = 1.8$

298. $p - 3.6 = 1.7$

299. $x - 0.4 = -3.9$

300. $y - 0.6 = -4.5$

301. $j - 1.82 = -6.5$

302. $k - 3.19 = -4.6$

303. $m - 0.25 = -1.67$

304. $q - 0.47 = -1.53$

305. $0.5x = 3.5$

306. $0.4p = 9.2$

307. $-1.7c = 8.5$

308. $-2.9x = 5.8$

309. $-1.4p = -4.2$

310. $-2.8m = -8.4$

311. $-120 = 1.5q$

312. $-75 = 1.5y$

313. $0.24x = 4.8$

314. $0.18n = 5.4$

315. $-3.4z = -9.18$

316. $-2.7u = -9.72$

317. $\frac{a}{0.4} = -20$

318. $\frac{b}{0.3} = -9$

319. $\frac{x}{0.7} = -0.4$

320. $\frac{y}{0.8} = -0.7$

321. $\frac{p}{-5} = -1.65$

322. $\frac{q}{-4} = -5.92$

323. $\frac{r}{-1.2} = -6$

324. $\frac{s}{-1.5} = -3$

Mixed Practice

In the following exercises, solve the equation. Then check your solution.

325. $x - 5 = -11$

326. $-\frac{2}{5} = x + \frac{3}{4}$

327. $p + 8 = -2$

328. $p + \frac{2}{3} = \frac{1}{12}$

329. $-4.2m = -33.6$

330. $q + 9.5 = -14$

331. $q + \frac{5}{6} = \frac{1}{12}$

332. $\frac{8.6}{15} = -d$

333. $\frac{7}{8}m = \frac{1}{10}$